

Jeonghyun Lee

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EDUCATION

University of California, Berkeley

Class of 2023

Bachelor of Arts in Data Science, Domain Emphasis in Applied Mathematics & Modeling

Berkeley, CA

- **Relevant Coursework:** Data Structures, Artificial Intelligence, Machine Learning, Data Engineering, Data Mining & Analytics, Probability for Data Science, Linear Algebra, Nonlinear & Discrete Optimization, Object-Oriented Programming
- **Additional Coursework:** Modern Robotics @ Northwestern University

EXPERIENCE

Alignerr

May 2026 - Present

Software Engineer, Machine Learning - AI/ML Model Evaluation & Benchmark Design

- Designed domain-specific robotics and multi-robot ML benchmarks, defining task specifications, evaluation data, reference solutions, and measurable scoring criteria to assess frontier LLM reasoning, domain understanding, and instruction adherence.

Soongsil University

Aug 2024 - Sep 2024

AI Engineer - LLM-Based Document Processing & Workflow Automation

Seoul, South Korea

- Built an event-driven Notion/Google Drive → PDF → OpenAI API → Gmail workflow using Make and Google Apps Script, converting uploaded PDFs to text, extracting structured receipt fields, and automatically generating downstream email content and delivery.

Outlier

Jan 2024 - Aug 2024

AI Training Specialist - AI Code Generation & Software Engineering Evaluation

San Francisco, CA

- Evaluated and ranked LLM-generated software patches for real-world Pandas, Koalas, and PySpark issues, validating type compatibility, indexing behavior, edge cases, backward compatibility, and regression risk using structured technical rubrics.

Electronics and Telecommunications Research Institute (ETRI)

Jul 2023 - Aug 2023

Machine Learning Engineer Intern - Multi-Robot SLAM & RL

Daejeon, South Korea

- Built a ROS/Gazebo multi-robot exploration environment for 4 TurtleBot3 robots using gmapping, move_base, and map_merge, and modified C++ RRT frontier detectors and Python filter/assigner nodes.
- Enabled SLAM map generation during DQN training by expanding LiDAR observations from 24 to 360 readings and state size from 28 to 364, addressing 'Scan matching failed' errors caused by sparse 15-degree LiDAR sampling.
- Trained independent DQN agents for 3,000 episodes with online SLAM map merging; scaling from 1 to 4 agents raised 5-minute map coverage from ~51% to ~94%, and single-agent DQN reached ~78% at 10 minutes vs. ~64% for single-agent RRT.

Climformatics

Sep 2022 - Dec 2022

Machine Learning Engineer Intern - Climate Data & Simulation

Richmond, CA

- Processed multidimensional NCAR CESM2 NetCDF data on Sabalcore HPC using NCO, NCL, and shell/PBS workflows, running diagnostics across 11 Beaufort Gyre historical ensemble members and extracting sea-ice concentration, thickness, and albedo.
- Reduced sea-ice thickness prediction MSE by 58.4% (0.250→0.104) by benchmarking Linear Regression and Decision Trees before training a 20-tree Random Forest on 8 years of Arctic-region albedo and sea-ice features.

University of California, Berkeley D-Lab

Jan 2022 - May 2022

Software Engineer Intern - Historical Text Data Processing & Linked Data

Berkeley, CA

- Processed ~15K Drehem Sumerian texts using Python, Pandas, and Jupyter for structured historical-text and network analysis.
- Developed regex- and rule-based NLP pipelines to extract occupations, transaction roles, and family relationships, and contributed to the LOD Tablet Dictionary by mapping BDTNS metadata and identifiers across ORACC, CDLI, FactGrid, and Wikidata.

PROJECT EXPERIENCE

Humanoid Robot Reinforcement Learning

- Reproduced HumanoidBench on Unitree H1 in MuJoCo and benchmarked 4 RL algorithms — SAC, PPO, TD-MPC2, and DreamerV3 — across multiple seeds, comparing sample efficiency, convergence, and training stability.
- Co-authored the 'Humanoid Robot for Everyone' on WikiDocs, contributing technical content on humanoid robotics.

Automated AI News Intelligence Pipeline

- Built a 7-stage Python news pipeline with normalization, URL cleanup, freshness filtering, deduplication, provenance merging, content trimming, and pipeline statistics; orchestrated Gemini analysis and source-linked HTML briefings through Make.

LIKELION Hackathon

- Built and deployed a navigation web app using Google Maps/Directions APIs, MongoDB, HTML/CSS, and AWS, implementing route calculation, path optimization, responsive map interactions, and persistent route-data storage.

TECHNICAL SKILLS

Languages: Python, C/C++, Java, SQL, JavaScript, R, MATLAB, Swift

ML / Data: PyTorch, TensorFlow, Keras, scikit-learn, Pandas, NumPy, SciPy, Matplotlib, xarray

Robotics / Systems: ROS, Gazebo, RViz, Gmapping, move_base, MuJoCo, Isaac Lab, Linux, Docker, Git/GitHub, Shell

Data / Infrastructure: HPC, NetCDF, NCO, NCL, PBS, Jupyter, MongoDB, MySQL, Supabase

Backend / APIs / Automation: Flask, Node.js, OpenAI API, Google Apps Script, Gemini, Make, Google Maps/Directions API, AWS, FastAPI